Electromagnetism & Vector Calculus

Magnetic Field & Flux

A magnetic field is a region of space where magnetic force is experienced. Recall that magnetic fields can either be formed by permanent magnets or by electrical charges in motion; current.

We quantify magnetic flux (ϕ) in Webers, Wb, where 1 Wb of magnetic flux is defined as the flux produced by a solenoid of 1 turn, under an emf of 1V, when the flux is reduced to zero in one second. Flux is also represented with field lines, flowing from north to south poles.

Magnetic flux density (B) is a way to mathematically represent the magnetic field, in units Tesla, T. Mathematically, it is a vector tangent to the flux lines/ a vector field of tangent vectors to the flux lines

We can represent the magnetic flux as a surface integral over a surface A, of the dot product of the magnetic flux density with the area vector (normal vector of magnitude being same as), or in other words,

$$\Phi = \int_A B \cdot dA$$

Essentially, the idea is that for a small section dA, we see how much the magnetic field is “influenced” by the area vector, by how aligned the two are.

Hence, we integrate the dot product, defined as

$$B \cdot dA = |B||dA| \cos(\theta)$$

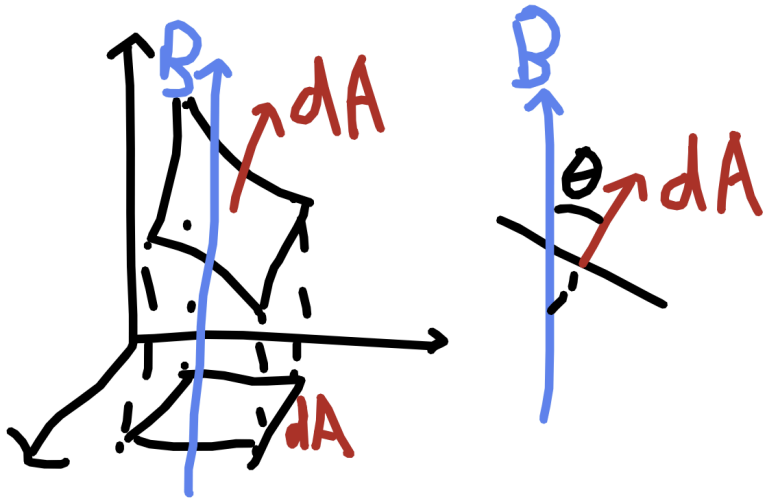
which also shows that when the magnetic flux lines are aligned to the area vector, theta is 0 thus the dot product is the product of the magnitudes, while when the vectors are perpendicular the dot product is 0.

Then, since $B \cdot dA$ only gives the magnetic flux over an area dA, we perform a surface integral.

In the case that B is constant, we can take out B to find that the magnetic flux is just

$$\Phi = B \cdot \int_A dA = B \cdot A = |B||A|\cos(\theta)$$

where B and A are the magnitudes of the magnetic flux density and area respectively, and theta is the angle between the vectors.



Computationally, the dot product can be found as

$$B \cdot A = \sum B_i A_i$$

If we have a solenoid of N turns, the flux passing through the surface bounded by each turn links the coil, that means that the magnetic flux produced by each coil should “combine” together, meaning that they are linked.

For a solenoid with an iron core, we can assume the linkage is equal for each turn, hence the total linkage for N turns is

$$\lambda = N\Phi$$

where λ is also in Wb or Wb turns (same unit)

Do also recall that the magnetic flux direction due to current, or vice versa can be determined via the right hand grip rule.

Magnetic Forces

Recall Fleming's Left Hand Rule, that states that for current flowing perpendicular in a magnetic field, it will experience a force.

We can extend this to non perpendicular angles, thus the force experienced by a charge q moving along a velocity vector u in a magnetic field vector B is

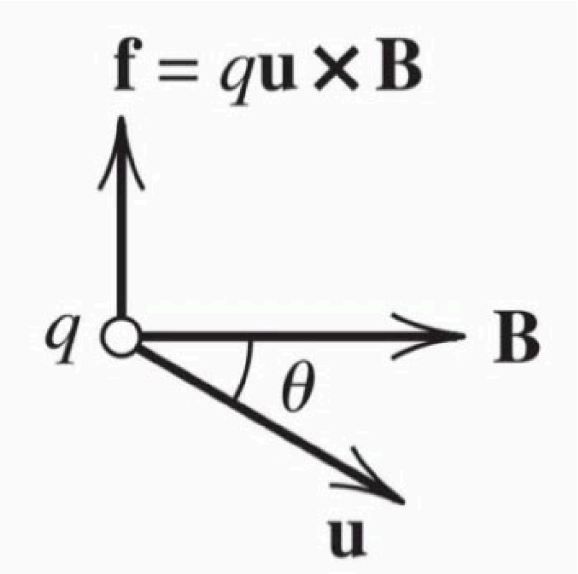
$$\mathbf{f} = q\mathbf{u} \times \mathbf{B}$$

where qu is actually current, except since it is about charges it will be negative to the current direction, if we were to consider a current carrying conductor.

The cross product vector will always be orthogonal to the two vectors, and oriented by the right hand rule (for $\mathbf{a} \times \mathbf{b}$, point index in a, middle in b, and thumb points to product direction)

And the magnitude of this vector is found as

$$|\mathbf{f}| = quB\sin(\theta)$$



which is from the cross product magnitude formula:

$$|\mathbf{a} \times \mathbf{b}| = |a||b|\sin(\theta)$$

If we were to compute the cross product in terms of basis vectors, it would be

$$\mathbf{f} = q\mathbf{u} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ qu_x & qu_y & qu_z \\ B_x & B_y & B_z \end{vmatrix}$$

which can be computed with Sarrus Rule.

Similarly, for a current carrying wire, we can find the force on an incremental length of wire as

$$d\mathbf{f} = id\mathbf{l} \times \mathbf{B}$$

which in the special case of a straight wire of length l,

$$f = ilB\sin(\theta)$$

The direction can be determined with Fleming's left hand rule or the right hand cross product rule.

Faraday's Law & Lenz's Law

Both these laws talk about the current/voltage produced from electromagnetic induction.

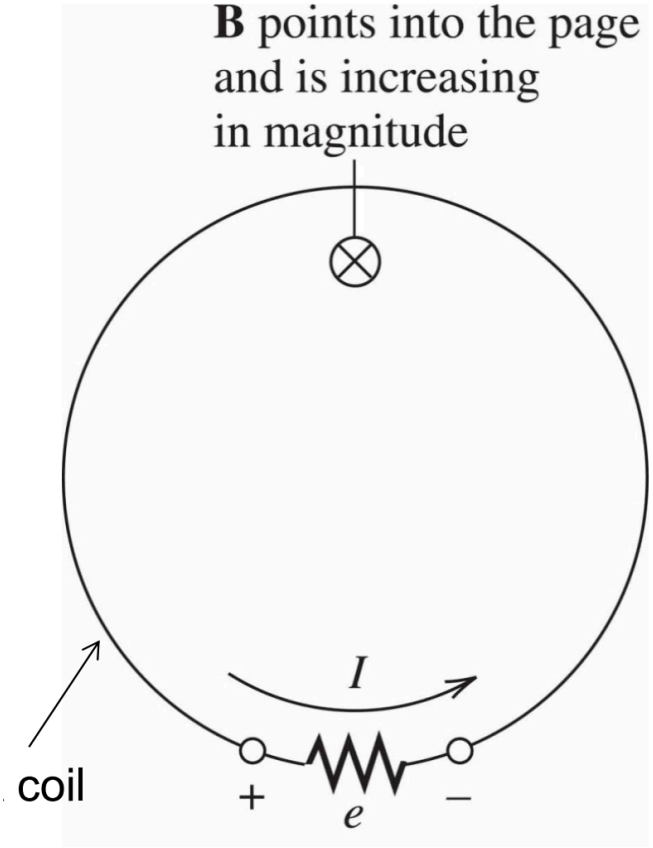
Faraday's Law states the a voltage e is induced in a coil due to a changing magnetic flux, or more specifically,

$$e = \frac{d\lambda}{dt} = N\frac{d\Phi}{dt}$$

Hence, this also means that the voltage can be induced by either the magnetic field changing itself, or the coil moving relative to the field.

Lenz's law states that the polarisation of the induced voltage is such that the voltage produces a current which opposes the changes in the flux linkages.

For example, if a changing magnetic field B passes perpendicularly through a coil into the page:



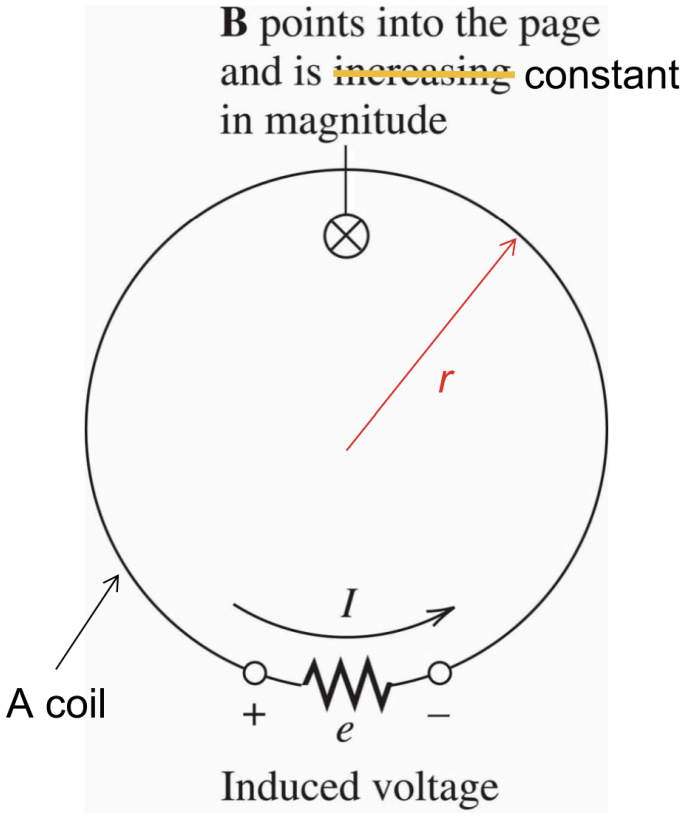
Faraday's Law states that a voltage will be induced proportional to the change, while Lenz's law states that the voltage induced will cause an anti clockwise current through some resistance R, producing a field out of the page, opposing the original field B.

In this case, the induced voltage can be found as

$$e = \frac{d\lambda}{dt} = \frac{d}{dt}(N\Phi) = \frac{d}{dt}(N(BA)) = NA\frac{dB}{dt}$$

Other than a changing magnetic flux, a voltage can also be induced by changing the size of the coil under a constant magnetic flux

Consider a constant magnetic field into the page, with a coil of radius r, expanding by Δr over a time Δt

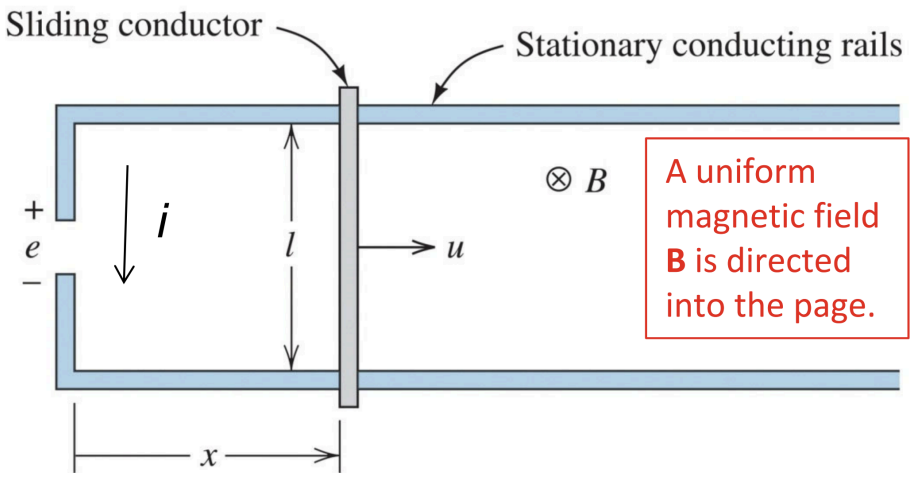


In this case, the induced voltage can be determined as

$$
\begin{aligned}
e &= \frac{d}{dt}(NBA) \\
&= NB\frac{d}{dt}(\pi r^2) \\
&= \pi NB\frac{d}{dt}(r^2) \\
&= 2\pi NB\frac{dr}{dt} \\
&= 2\pi NB\frac{\Delta r}{\Delta t}
\end{aligned}
$$

and to find the direction of the voltage/current, we know that the magnetic field produced should oppose the original B, thus it should be out of the page. Hence, the current should flow anticlockwise and the voltage induced should follow this convention.

Another example of changing area in a constant field is shown:



If the sliding conductor is pulled along, the voltage induced is

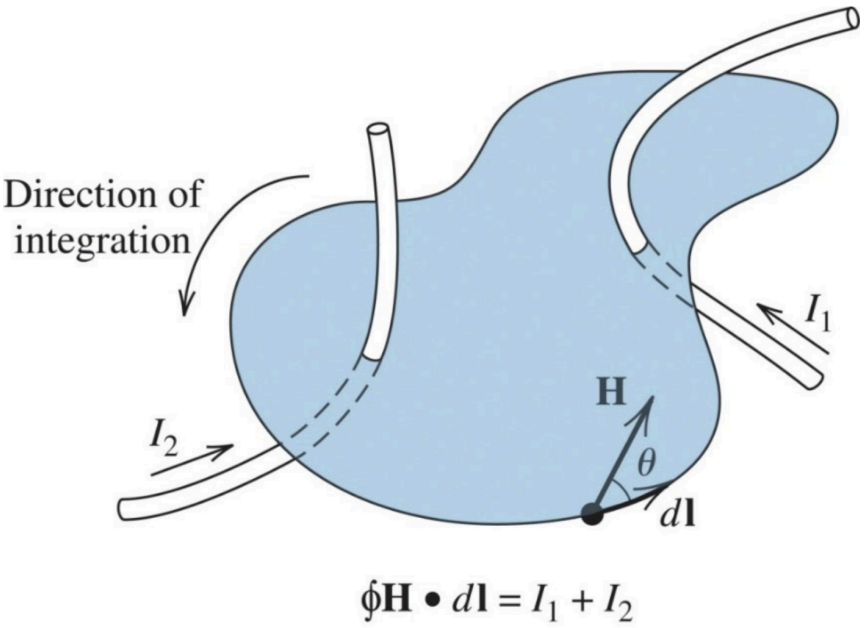
$$
\begin{aligned}
e &= \frac{d}{dt}(NBA) \\
&= B\frac{dA}{dt}\ [N=1] \\
&= B\frac{d}{dt}(lx) \\
&= Bl\frac{dx}{dt} \\
&= Blu
\end{aligned}
$$

In this case, there is a voltage induced however current does not flow as it is an open circuit.

Ampere's Law

Ampere's Law states that the line integral of the magnetic field intensity (H), in units ampere-meters, around a closed path is equal to the algebraic sum of the currents flowing through the area. In other words,

$$\oint \mathbf{H} \cdot d\mathbf{l} = \sum i$$



The magnetic field intensity, H can be found from

$$B = \mu H$$

The idea is that H is the magnetising force used to “create” the magnetic field, but the coefficient μ, the magnetic permeability, represents how easily the magnetic field can be setup in the material.

∗∗Recall the permeability of a vacuum/free space:

$$\mu_0 = 4\pi \times 10^{-7} Wb/Am$$

and the relative permeability of a material:

$$\mu_r = \frac{\mu}{\mu_0}$$

Thus, we could rewrite Ampere’s Law as

$$\oint \frac{1}{\mu}\mathbf{B} \cdot d\mathbf{l} = \sum i$$
$$\frac{1}{\mu}\oint \mathbf{B} \cdot d\mathbf{l} = \sum i$$
$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu \sum i$$

Firstly, $\mathbf{B} \cdot d\mathbf{l}$ is a measure of how aligned the magnetic flux vector B is with the tangential vector dl at a point along the path. When we sum them all up with the closed integral, it represents the total magnetic flux due to the sum of the currents i, by a factor of the permeability.

For the sum of currents, to determine their sign we see if the current contributes a magnetic flux in the same direction as the direction of the integration, by the right hand grip rule, then that means it is a positive current.

Say we had the case where H is constant H is always tangential to the path, that means the angle between them is 0 and the closed loop integral can be evaluated easily as

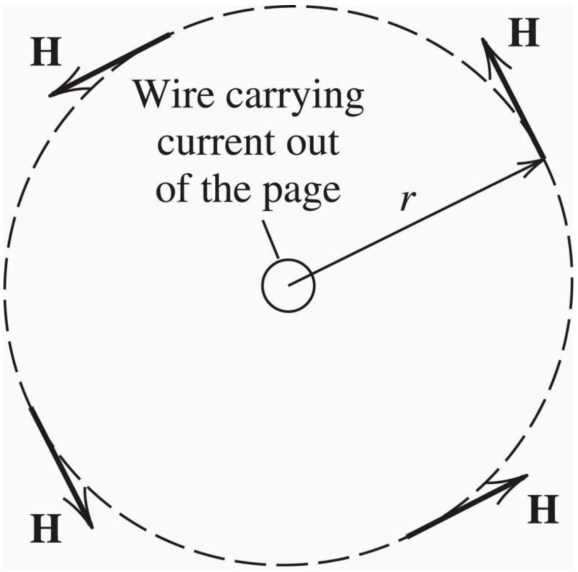
$$H \oint d\mathbf{l} = \sum i$$
$$Hl = \sum i$$

or

$$Bl = \mu \sum i$$

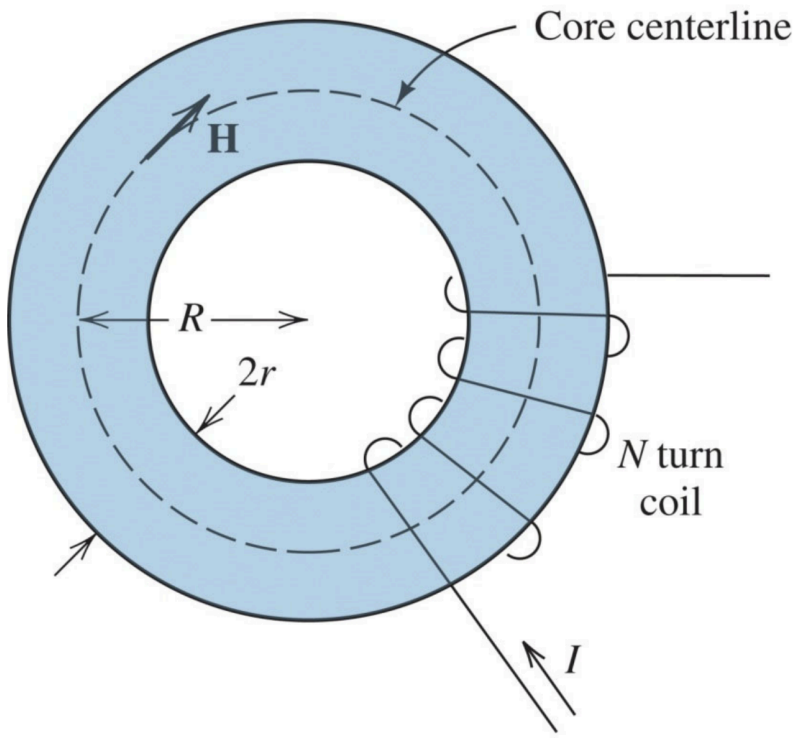
where l is the total length of the closed loop.

For example, consider the following:



A wire carries current out of the page and produces a magnetic field intensity around a circular flux line like so. To find the magnetic flux density in this loop, we know H is constant and tangential and thus

$$Bl = \mu I$$
$$B(2\pi r) = \mu I$$
$$B = \frac{\mu I}{2\pi r}$$



In another case, say we had to find the magnetic flux density of the toroidal core's centerline, and the magnetic flux and the flux linkage of the coil area

Again, it is a circle and by the right hand grip rule of the coil, magnetic field strength H is tangential to the circle. Hence, we can use $Hl = \sum i$

Notice how there are N number of conductors cutting into the region defined by the centerline. They all flow in the same direction and contribute to the direction of H. Hence,

$$Hl = \sum i$$
$$\frac{B}{\mu}(2\pi r) = Ni$$
$$B = \frac{\mu NI}{2\pi R}$$

To find the magnetic flux density, we first need to assume that B is uniform throughout (which is true when R is much greater than r)

This lets us assume that B is constant, and thus since B points perpendicularly to the coil areas,

$$\Phi = BA$$
$$\Phi = \frac{\mu NI}{2\cancel{\pi} R} \times \cancel{\pi} r^2$$
$$\Phi = \frac{\mu NIr^2}{2R}$$

and hence the flux linkage is

$$\lambda = N\Phi = \frac{\mu N^2 Ir^2}{2R}$$

Inductance

Recall inductors, and inductance being a measure of how much energy it can store in a magnetic field. Inductance can actually be defined as

$$L = \frac{\lambda}{i}$$

which also shows that $1H = 1Wb/A$, and that inductance can be thought of as the amount of magnetic flux produced per unit current.

If we were to make the flux linkage the subject, we get that $\lambda = Li$. If we apply this to a solenoid/inductor that has current flowing through, a voltage is induced by Faraday's Law:

$$v = \frac{d\lambda}{dt} = L\frac{di}{dt}$$

which matches with the characteristic equation of inductors.

The inductance of different inductor geometries can be found too, by first using $\lambda = N\Phi$ to get

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$$L = \frac{N\Phi}{i}$$
$$L = \frac{NBA}{i}$$

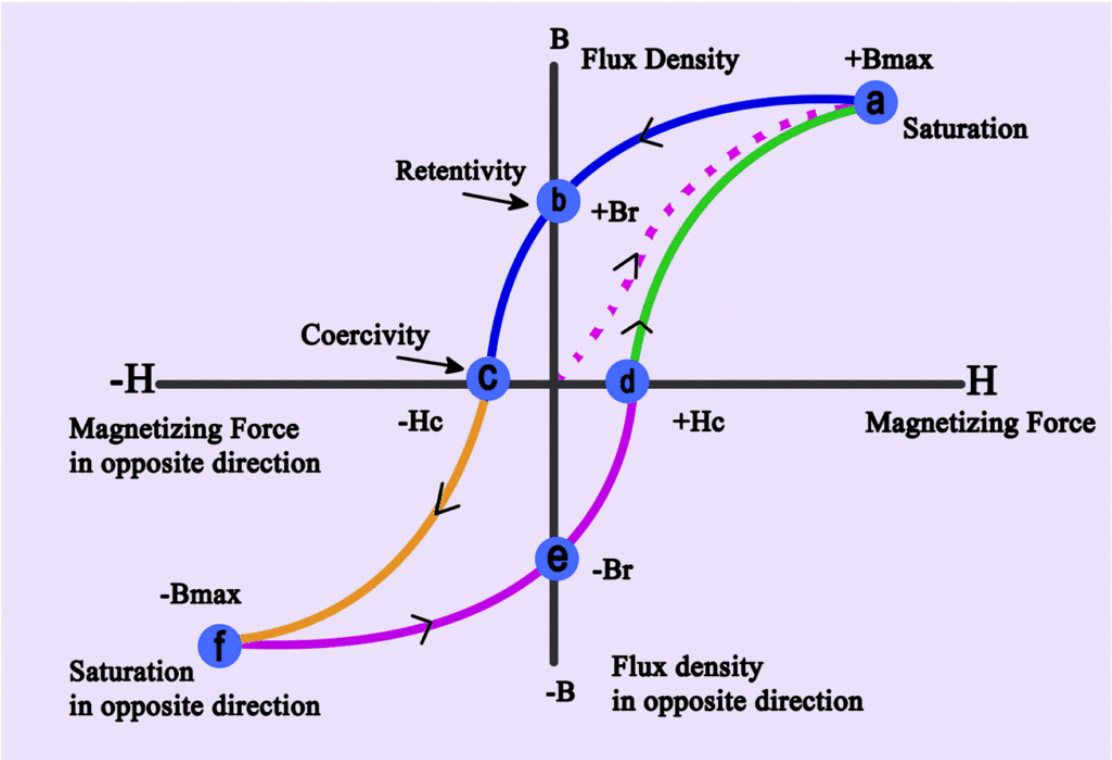
where B and A can be found with Amperes Law of different coils.

Magnetic Materials

The equation $B = \mu H$ may look like it implies B is directly proportional to H, however it is actually not linear due to the hystersis curve:

Magnetic Hysteresis & Hysteresis Curve

Hysteresis is when the magnetisation of a material (B) lags from the applied magnetic force(H), which mainly depends on the input current



The Hysteresis/B-H curve shows the relationship between the two values, where the positive/negative polarity of B represents a swap in magnetic polarity, while the positive/negative polarity of H represents the direction of input current and hence the direction of the magnetising force.

The curve represents a setup with a solenoid wrapped around an originally unmagnetised magnetic core, where the current can be varied. and the magnetic flux of the core is measured.

There are 7 phases of this diagram:

Phase 1: Origin to A

Initially, as the core is not magnetised and no current is applied, no magnetising force is applied and hence no magnetic flux is formed. (i.e. B=0,H=0 at origin)

As a current & magnetising force is applied, the flux density increases proportionally up to a point a, where it starts to level out. We call this saturation, as we cannot increase the flux density anymore even if we add more magnetising force. Thus, we also call point a the saturation point/maximum point

Phase 2: a to b

As H decreases, B will also decrease but will decrease at a slower rate and not fall back to 0Wb(which we refer to B_r) This means that even though we had removed the magnetising force, we "permanently" magnetised our core with the solenoid, as some of the domains are still aligned, however with lesser flux density than what we had put in.

This effect of staying magnetised even after an external magnetic force is removed is called retentivity (retention of flux density)

Phase 3: b to c

Now, we then change the direction of current, so that H is negative (opp direction)

This causes the flux density to decrease more to point c, aka when flux density is 0.

This can be thought of as the opposite magnetic force messing up the domains, causing them to no longer be aligned together, thus having B=0

The point where c intersects on the H axis is the coercive force, $-H_c$, which is the magnetising force required to remove the retentivity and make B=0.

Phase 4: c to f

If we further increase current and H in the opposite direction, the flux density will also increase similarly in phase 1, however in the opposite direction. B will then continue to increase in the negative

direction until it reaches saturation again, at the same magnitude in phase 1. Similar properties of saturation apply here.

Phase 5: f to e

if we decrease H back to 0 again, B will also decrease at a slower rate, reaching $-B_r$.

Phase 6: e to d

increasing H back to the positive direction causes the flux density to reach 0, where the curve cuts at $+H_c$.

Phase 7: d to a

Similar to phase 1, as we increase H again however from a non 0 value, B will reach the saturation point.

Phases 2 to 7 will then repeat for when H changes.

Note that different materials will have differently shapes B-H curves

From this curve we now know of 3 properties of magnetisation, permeability, retentivity and coercivity

Permeability: How easily a magnetic field can be setup

Retentivity: How well a magnetic field can be maintained/retained (w/o an external field)

Coercivity: How hard it is to demagnetise a magnetic field.

high retentivity and coercivity is useful for permanent magnets, where you want to maintain the magnetic field.

High permeability and low coercivity is useful for electromagnetic alarms and circuit breakers, where a core needs to be rapidly magnetised and demagnetised.

High coercivity is also useful for credit cards, so that information is harder to erase, while low coercivity is useful for single trip train cards, where information can be easily erased to be reused again

Hysteresis also causes energy loss, called core loss, which is proportional to the area of the hysteresis loop.